Assignment 3:
Set A:

1) Design following Android Portrait and Landscape Screen Layout to add a border.

To add a border to any layout:

- We are creating separate xml file for border. Required attributes of borders are set in the xml.
- Apply the designed border to your layout in activity_main.xml file.

1. Create Android Project

- New Project → **Empty view Activity**
- Language: **Java**
- Create 2 values xml files with names:
- border.xml and newborder.xml

Border.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    android:shape="rectangle">
    <!-- Border -->
    <stroke
        android:width="8dp"
        android:color="@color/black" />
    <!-- Optional background -->
    <solid
android:color="@android:color/holo_blue_dark" />
    <!-- Optional rounded corners -->
    <corners android:radius="8dp" />

    <!-- Optional padding inside border -->
    <padding
        android:left="8dp"
        android:top="8dp"
        android:right="8dp"
        android:bottom="8dp" />
</shape>
```

Newborder.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape
xmlns:android="http://schemas.android.com/apk/res/andro
id"
```

```
    android:shape="rectangle">
    <!-- Border -->
    <stroke
        android:width="8dp"
        android:color="@color/white" />
    <!-- Optional background -->
    <solid
android:color="@color/material_dynamic_neutral50" />
    <!-- Optional rounded corners -->
    <corners android:radius="8dp" />
    <!-- Optional padding inside border -->
    <padding
        android:left="8dp"
        android:top="8dp"
        android:right="8dp"
        android:bottom="8dp" />
</shape>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="30dp"
    android:background="@drawable/border">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="500dp"
        android:background="@drawable/newborder"
        android:gravity="center"
        android:text="Hello World!"
        android:textSize="28sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output: Shows different borders applied to constraint Layout and TextView each.

Set A: 2) Create a simple calculator to perform appropriate arithmetic operations.

1. Create Android Project

- New Project → **Empty view Activity**
- Language: **Java**
- Name it: Calculator

2. Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <TextView
        android:id="@+id/etNum"
        android:layout_width="match_parent"
        android:layout_height="50dp"
        android:layout_marginTop="10dp"
        android:hint="0"
        android:textAlignment="viewEnd"
        android:textSize="30dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center"
        android:layout_marginTop="20dp">

        <Button
            android:id="@+id/btn1"
            android:text="1"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
```

```
        <Button
            android:id="@+id/btn2"
            android:text="2"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btn3"
            android:text="3"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btnplus"
            android:text="+"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>

    </LinearLayout>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">

        <Button
            android:id="@+id/btn4"
            android:text="4"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>

        <Button
            android:id="@+id/btn5"
            android:text="5"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
```

```
        <Button
            android:id="@+id/btn6"
            android:text="6"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btnminus"
            android:text="-"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>

    </LinearLayout>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">

        <Button
            android:id="@+id/btn7"
            android:text="7"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>

        <Button
            android:id="@+id/btn8"
            android:text="8"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btn9"
            android:text="9"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
```

```
            android:id="@+id/btnmultiply"
            android:text="x"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
    </LinearLayout>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">
        <Button
            android:id="@+id/btnClear"
            android:text="Clr"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btn0"
            android:text="0"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btndevide"
            android:text="/"
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
        <Button
            android:id="@+id/btnequals"
            android:text="="
            android:textSize="30dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1"/>
    </LinearLayout>
</LinearLayout>
```

3. MainActivity.java

```
package com.example.calculator;

import static com.example.calculator.R.id.*;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    Button
b1,b2,b3,b4,b5,b6,b7,b8,b9,b0,bplus,bminus,bmulti,bdevi
de,btnClear,btnequals;
    TextView num;
    @Override
    protected void onCreate(Bundle savedInstanceState)
{
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        b1=findViewById(R.id.btn1);
        b2=findViewById(R.id.btn2);
        b3=findViewById(R.id.btn3);
        b4=findViewById(R.id.btn4);
        b5=findViewById(R.id.btn5);
        b6=findViewById(R.id.btn6);
        b7=findViewById(R.id.btn7);
        b8=findViewById(R.id.btn8);
        b9=findViewById(R.id.btn9);
        b0=findViewById(R.id.btn0);
        bplus=findViewById(R.id.btnplus);
        bminus=findViewById(R.id.btnminus);
        bmulti=findViewById(R.id.btnmultiply);
        bdevide=findViewById(R.id.btndevide);
        btnequals=findViewById(R.id.btnequals);
        btnClear=findViewById(R.id.btnClear);
```

```
        num=findViewById(R.id.etNum);

        b1.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("1");
            }
        });
        b2.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("2");
            }
        });
        b3.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("3");
            }
        });
        b4.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("4");
            }
        });
        b5.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("5");
            }
        });
        b6.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("6");
            }
        });
        b7.setOnClickListener(new
```

```
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("7");
            }
        });
        b8.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("8");
            }
        });
        b9.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("9");
            }
        });
        b0.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("0");
            }
        });
        bplus.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("+");
            }
        });
        btnClear.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.setText("");
            }
        });
        bmulti.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```
                num.append("*");
            }
        });
        bminus.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("-");
            }
        });
        bdevide.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                num.append("/");
            }
        });

        btnequals.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input=num.getText().toString();
                int output=0;
                int num1 = 0, num2 = 0;
                char operator = ' ';

                // Find the operator
                for (int i = 0; i < input.length();
i++) {
                    char ch = input.charAt(i);
                    if (ch == '+' || ch == '-' || ch ==
'*' || ch == '/') {
                        operator = ch;
                        num1 =
Integer.parseInt(input.substring(0, i));
                        num2 =
Integer.parseInt(input.substring(i + 1));
                        break;
                    }
                }

                switch (operator) {
                    case '+':
                         output=num1 + num2;
```

```
                                break;
                            case '-':
                                output=num1-num2;
                                break;
                            case '*':
                                output=num1*num2;
                                break;
                            case '/':
                                if (num2 == 0) {
                                    num.setText("Division by
zero");
                                }
                                output=num1 / num2;
                                break;
                            default:
                                num.setText("Invalid input");
                                break;
                        }
                        num.setText(String.valueOf(output));
                    }
                });
            }
}
```

Output:

This code will generate calculator with all basic functions.

Design a Screens Using Intents to insert customer contact details.

Flow

1. **MainActivity** → has **"Add Contact"** button
2. Clicking button opens **AddContactActivity**
3. User enters contact details
4. After clicking **Save**, a **ContactDetailsActivity** opens and shows the contact details (short view)

---

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnAddContact"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add Contact" />
</LinearLayout>
```

MainActivity.java

```
package com.example.contactapp;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnAddContact;

    @Override
    protected void onCreate(Bundle savedInstanceState)
```

```
{
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnAddContact =
findViewById(R.id.btnAddContact);

        btnAddContact.setOnClickListener(v -> {
            Intent intent = new
Intent(MainActivity.this, AddContactActivity.class);
            startActivity(intent);
        });
    }
}
```

activity_add_contact.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">
    <ImageView
        android:id="@+id/imgContact"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@drawable/ic_launcher_foreground"
        android:layout_gravity="center"
        android:scaleType="centerCrop"/>

    <Button
        android:id="@+id/btnChooseImage"
        android:text="Choose Image"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
    <EditText
        android:id="@+id/etName"
        android:hint="Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
```

```
        android:id="@+id/etPhone"
        android:hint="Phone"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etEmail"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnSave"
        android:text="Save Contact"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

AddContactActivity.java

```
package com.example.contactapp;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.MediaStore;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ImageView;

import androidx.appcompat.app.AppCompatActivity;

public class AddContactActivity extends
AppCompatActivity {

    EditText etName, etPhone, etEmail;
    Button btnSave, btnChooseImage;
    ImageView imgContact;
    Uri imageUri;

    @Override
    protected void onCreate(Bundle savedInstanceState)
{
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_add_contact);

        etName = findViewById(R.id.etName);
        etPhone = findViewById(R.id.etPhone);
        etEmail = findViewById(R.id.etEmail);
        btnSave = findViewById(R.id.btnSave);
        btnChooseImage =
findViewById(R.id.btnChooseImage);
        imgContact = findViewById(R.id.imgContact);

        btnChooseImage.setOnClickListener(v -> {
            Intent intent = new
Intent(Intent.ACTION_PICK,

MediaStore.Images.Media.EXTERNAL_CONTENT_URI);
        });

        btnSave.setOnClickListener(v -> {
            Intent intent = new
Intent(AddContactActivity.this,
contactDetailActivity.class);
            intent.putExtra("name",
etName.getText().toString());
            intent.putExtra("phone",
etPhone.getText().toString());
            intent.putExtra("email",
etEmail.getText().toString());
            intent.putExtra("image",
imageUri.toString());
            startActivity(intent);
        });
    }

    @Override
    protected void onActivityResult(int requestCode,
int resultCode, Intent data) {
        super.onActivityResult(requestCode, resultCode,
data);

        if (requestCode == 100 && resultCode ==
RESULT_OK) {
            imageUri = data.getData();
            imgContact.setImageURI(imageUri);
        }
```

```
    }
}
```

activity_contact_detail.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    android:orientation="vertical">
    <ImageView
        android:id="@+id/imgContactDetails"
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:layout_gravity="center"
        android:scaleType="centerCrop"/>

    <TextView
        android:id="@+id/tvDetails"
        android:textSize="18sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

contactDetailActivity.java

```
package com.example.contactapp;

import android.net.Uri;
import android.os.Bundle;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class contactDetailActivity extends
AppCompatActivity {

    TextView tvDetails;
    ImageView imgContactDetails;
    @Override
    protected void onCreate(Bundle savedInstanceState)
{
        super.onCreate(savedInstanceState);
```

```
setContentView(R.layout.activity_contact_detail);

        tvDetails = findViewById(R.id.tvDetails);

        imgContactDetails =
findViewById(R.id.imgContactDetails);
        String name =
getIntent().getStringExtra("name");
        String phone =
getIntent().getStringExtra("phone");
        String email =
getIntent().getStringExtra("email");

        String imageUri =
getIntent().getStringExtra("image");

imgContactDetails.setImageURI(Uri.parse(imageUri));
        tvDetails.setText(
                "Name: " + name + "\n" +
                        "Phone: " + phone + "\n" +
                        "Email: " + email
        );
    }
}
```

Set C: Create the following layout which is changing android spinner text.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/tvResult"
        android:text="Selected item will appear here"
        android:textSize="40dp"
        android:layout_marginTop="80dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.spinner;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.view.View;
import android.widget.AdapterView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Spinner spinner;
    TextView tvResult;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState)
{
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        spinner = findViewById(R.id.spinner);
        tvResult = findViewById(R.id.tvResult);

 String[] items = {"Filter", "Espresso", "Americano",
"Cappuccino",”Latte”,”Mocha”,”Cortado”,”Flat
White”,”Macchiato”};

        ArrayAdapter<String> adapter = new
ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_item,
                items
        );

adapter.setDropDownViewResource(android.R.layout.simple
_spinner_dropdown_item);
        spinner.setAdapter(adapter);

        spinner.setOnItemSelectedListener(new
AdapterView.OnItemSelectedListener() {
            @Override
            public void onItemSelected(AdapterView<?>
parent, View view, int position, long id) {
                tvResult.setText("Selected: " +
items[position]);
            }

            @Override
            public void
onNothingSelected(AdapterView<?> parent) {
                tvResult.setText("Nothing selected");
            }
        });
    }
}
```